

Lecture Notes

Group 4

Course: AELL 101

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1 Filters

- A Filter is an AC circuit that **separates** some frequencies from others within mixed-frequency signals. The circuits are designed to allow certain frequencies to pass while attenuating (weaken) others.

- Uses of Filters

1. **Noise Reduction:** To remove undesired noise or interference from a signal. This is essential for applications requiring a clean signal, such as audio processing and communication systems.
2. **Signal Smoothing:** To reduce fluctuations in the signal and produce an output that is steady and continuous.

1. Low- Pass Filter

- (a) Allows low-frequency signals to **pass** while blocking frequencies higher than the cut-off frequencies.

- (b) There are two types of Low- Pass Filters

- i. **Inductive Low- Pass Filters**

The inductor's impedance increases with increasing frequency. This high impedance in series blocks the high- frequency signals to pass through the load.

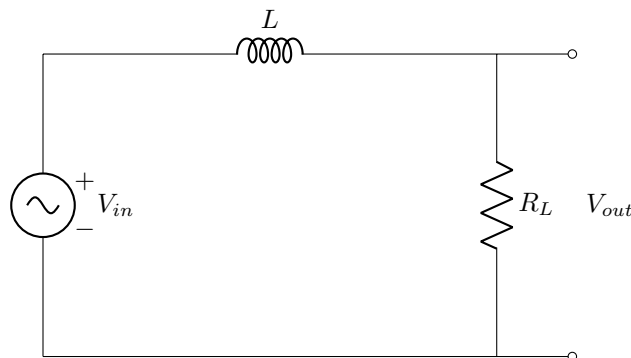


Figure 1: Low- Pass Inductive Filter

ii. Capacitive Low- Pass Filters

- As frequency increases, the capacitor's impedance decreases. In parallel with the load resistance.
- The capacitor acts like a short circuit for high-frequency signals, diverting them away from the load.
- Since the capacitor is effectively shunting (bypassing) the high-frequency signal, less current flows through the load resistance.
- If less current flows through the load resistance, the voltage drop across it remains significant. This results in the voltage across the load resistor remaining close to the input voltage.

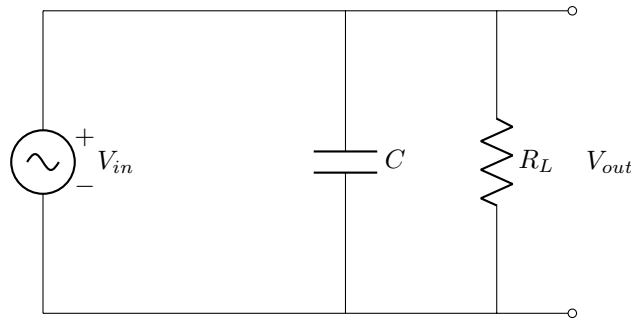


Figure 2: Low- Pass Capacitive Filter

2. High- Pass Filter

- Allows high-frequency signals to pass while blocking frequencies lower than the cut-off frequencies.
- There are two types of High- Pass Filters

i. Inductive High- Pass Filters

- As the frequency increases, the impedance also increases in an inductor.
- At very high frequencies, the impedance becomes so large that the inductor behaves almost like an open circuit, allowing very little current to pass through it.
- The resistor receives a significant voltage, meaning high frequencies are allowed to pass through.

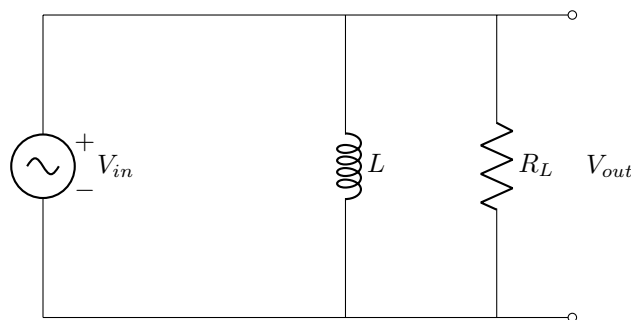


Figure 3: High- Pass Inductive Filter

ii. Capacitive High- Pass Filters

The capacitor's impedance increases with decreasing frequency. This high impedance in series blocks low-frequency signals from getting to load.

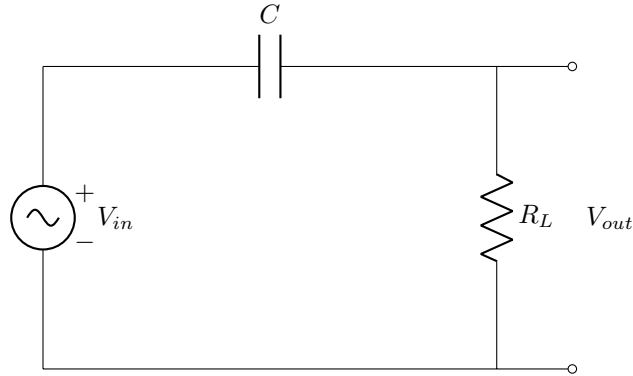


Figure 4: High- Pass Capacitive Filter

- Capacitive filters are usually **preferred** over inductive ones because capacitors are almost entirely reactive (they have very little resistance). Inductors often have internal resistances due to the long wire coils and energy losses in their core material, making them less efficient.

- Cut-off Frequency

$$f_c = \frac{1}{2\pi RC}$$

2 AC Vs DC

2.1 Differences Between AC and DC

Table 1: Differences between AC and DC

Feature	Alternating Current (AC)	Direct Current (DC)
Direction	Changes periodically	Flows in one direction
Frequency	Typically 50Hz or 60Hz	Zero (constant)
Energy Transmission	Efficient for long distances	Inefficient for long distances
Voltage Conversion	Easily transformed using transformers	Requires complex circuits
Common Uses	Power grids, home appliances	Batteries, electronic devices

2.2 Advantages of AC source

- **Efficient Power Transmission** – AC can be transmitted over long distances with minimal power loss using transformers.
- **Easier Voltage Transformation** – AC voltage can be easily stepped up or down using transformers.
- **Widely Used in Homes and Industries** – Most household appliances, industrial equipment, and power grids use AC.
- **Less Maintenance Required** – AC motors and generators have fewer components subject to wear and tear.

3 3- Phase AC

3.1 Single Phase AC

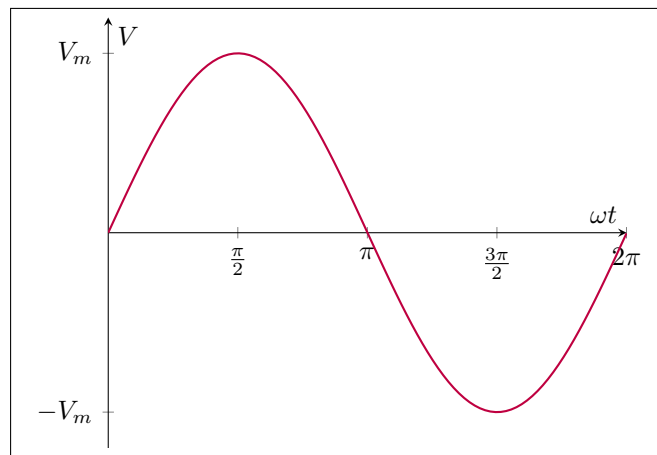


Figure 5: Single Phase AC Waveform

$$V = V_m \sin \omega t$$

$$V = V_{rms} \angle \phi^\circ$$

A single phase AC source operates with only **one alternating** voltage waveform.

3.2 Three Phase AC

A balanced three-phase source consists of three single phase sources with equal magnitude and 120° phase shift between each source.

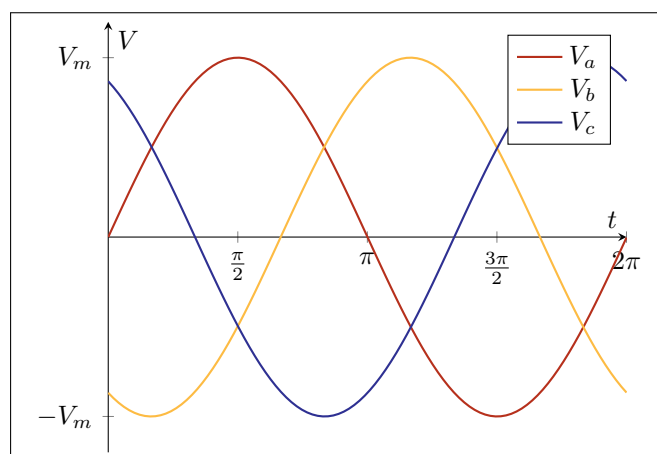


Figure 6: 3 Phase AC Waveform

$$V_a = \frac{V_m}{\sqrt{2}} \angle 0^\circ \quad (1)$$

$$V_b = \frac{V_m}{\sqrt{2}} \angle -120^\circ \quad (2)$$

$$V_c = \frac{V_m}{\sqrt{2}} \angle 120^\circ \quad (3)$$

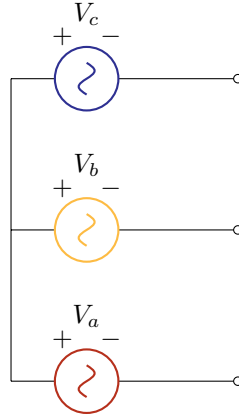


Figure 7: 3 - ϕ Source

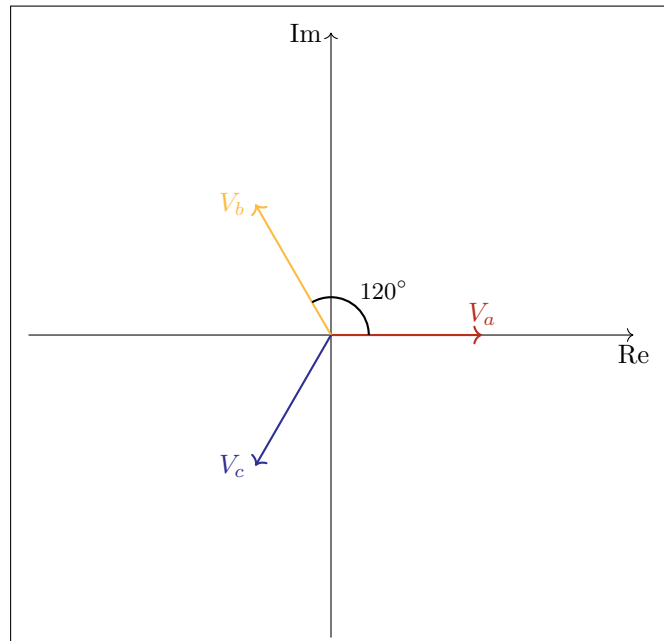


Figure 8: Phasor Diagram of 3 Phase AC Source

$$\begin{aligned}
V_{\text{net}} &= V_a + V_b + V_c \\
&= \frac{V_m}{\sqrt{2}} \angle 0^\circ + \frac{V_m}{\sqrt{2}} \angle -120^\circ + \frac{V_m}{\sqrt{2}} \angle 120^\circ \\
&= \frac{V_m}{\sqrt{2}} + \left(\frac{V_m}{\sqrt{2}} \cos(-120^\circ) + j \frac{V_m}{\sqrt{2}} \sin(-120^\circ) \right) \\
&\quad + \left(\frac{V_m}{\sqrt{2}} \cos(120^\circ) + j \frac{V_m}{\sqrt{2}} \sin(120^\circ) \right) \\
&= \frac{V_m}{\sqrt{2}} - \frac{V_m}{\sqrt{2}} \\
V_{\text{net}} &= 0
\end{aligned}$$

When $V_{\text{net}} = 0$ the system is called a balanced three phase system

4 Power in 3- Phase AC

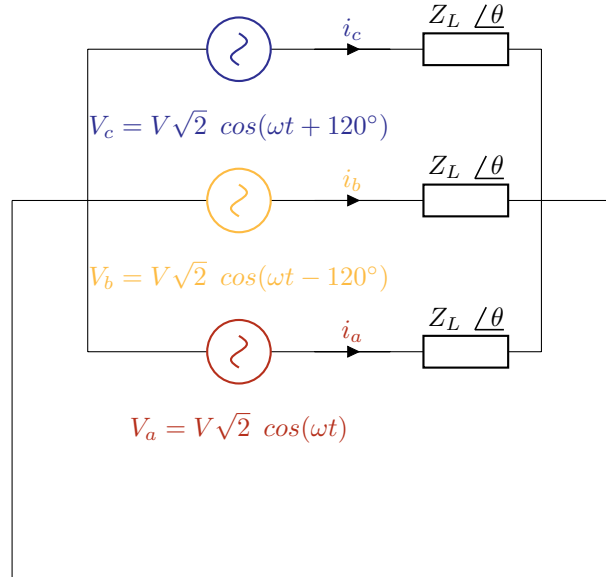


Figure 9: 3 - ϕ Source and Identical Loads

$$P(t) = V_{\text{rms}} I_{\text{rms}} (\cos(\phi) + \cos(2\omega t + \phi)) \quad (4)$$

$$\begin{aligned}
P_{\text{net}} &= P_a + P_b + P_c \\
&= V_a i_a + V_b i_b + V_c i_c \\
&= \sqrt{2}V \cos(\omega t) \sqrt{2}I \cos(\omega t - \theta) \\
&\quad + \sqrt{2}V \cos(\omega t - 120^\circ) \sqrt{2}I \cos(\omega t - \theta - 120^\circ) \\
&\quad + \sqrt{2}V \cos(\omega t + 120^\circ) \sqrt{2}I \cos(\omega t - \theta + 120^\circ) \\
&= 2VI \left[\cos(\omega t) \cos(\omega t - \theta) \right. \\
&\quad \left. + \cos(\omega t - 120^\circ) \cos(\omega t - \theta - 120^\circ) \right. \\
&\quad \left. + \cos(\omega t + 120^\circ) \cos(\omega t - \theta + 120^\circ) \right] \\
\Rightarrow \cos A \cos B &= \frac{1}{2} [\cos(A + B) + \cos(A - B)] \\
P_{\text{net}} &= VI \left[3 \cos \theta + \cos(2\omega t - \theta) \right. \\
&\quad \left. + \cos(2\omega t - \theta - 240^\circ) + \cos(2\omega t - \theta + 240^\circ) \right] \\
\Rightarrow \cos(A + B) &= \cos A \cos B - \sin A \sin B \\
\Rightarrow \cos(A - B) &= \cos A \cos B + \sin A \sin B \\
P_{\text{net}} &= VI \left[3 \cos \theta + \cos(2\omega t - \theta) \right. \\
&\quad \left. + \cos(2\omega t - \theta) \cos 240^\circ + \sin(2\omega t - \theta) \sin 240^\circ \right. \\
&\quad \left. + \cos(2\omega t - \theta) \cos 240^\circ - \sin(2\omega t - \theta) \sin 240^\circ \right] \\
&= VI \left[3 \cos \theta + \cos(2\omega t - \theta) + 2 \left(-\frac{1}{2} \cos(2\omega t - \theta) \right) \right] \\
&= VI [3 \cos \theta]
\end{aligned}$$

The total instantaneous power in a balanced three-phase system is **constant**. Even though the instantaneous power of each phase is time-varying. This result is true for both Y- or Δ - connected load.

5 Source Connections

5.1 Y Connections

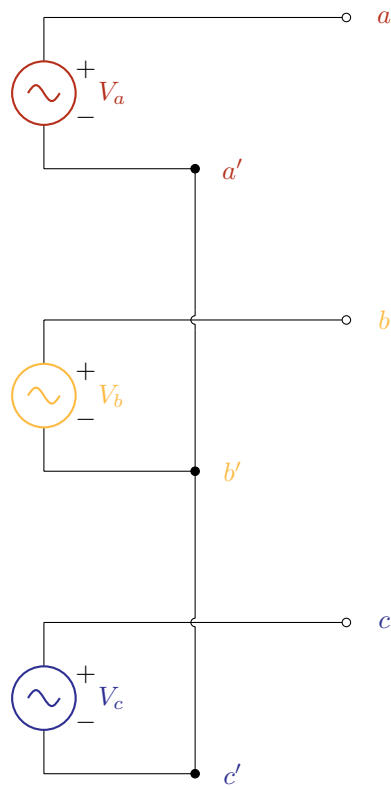


Figure 10: Y- Connection

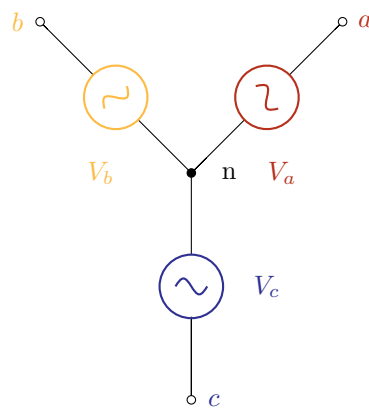


Figure 11: Y- Connection

5.2 Δ Connection

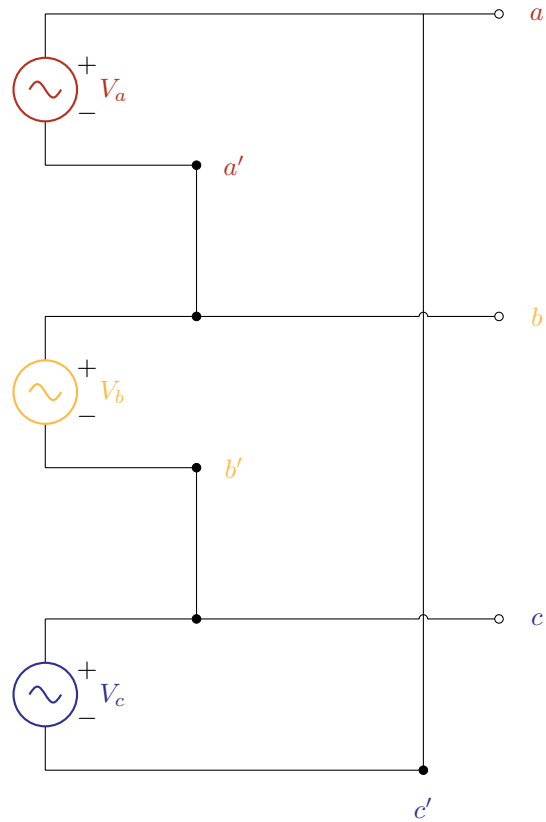


Figure 12: Δ - Connection

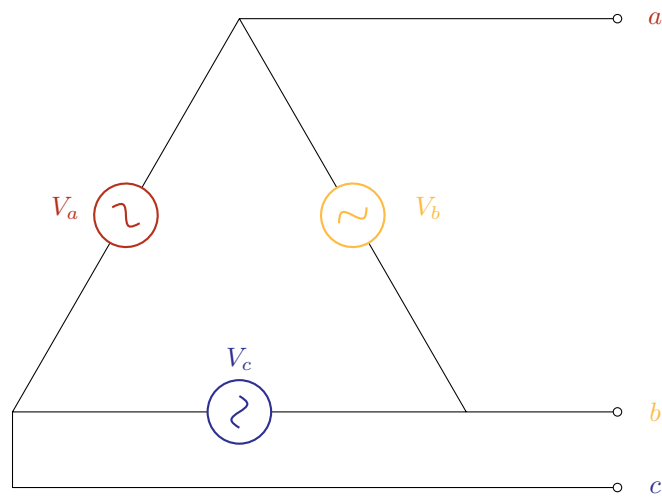


Figure 13: Δ - Connection

6 Three phase Quantities

6.1 Phase and Line Voltages

- Phase voltage (V_P) refers to the voltage measured across any one source or load in a balanced three-phase source or load.
- Line voltage (V_L) is the voltage measured between any two lines in a balanced three-phase system

6.2 Phase and Line Currents

- Phase current (I_P) is the current through any one component.
- Line current (I_L) is the current through any one line in a balanced three-phase source or load.

6.3 Phase and line voltages in Y connection

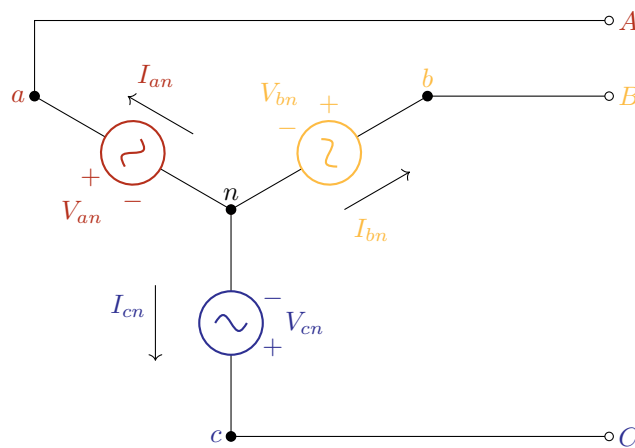


Figure 14: Three phase voltage source in Y connection

$$V_{an} = V \angle 0^\circ$$

$$V_{bn} = V \angle -120^\circ$$

$$V_{cn} = V \angle 120^\circ$$

Line voltage across the lines a and b is given by:

$$\begin{aligned}
 V_{ab} &= V_{an} - V_{bn} \\
 &= V\angle 0^\circ - V\angle -120^\circ \\
 &= V - (V \cos(-120) + jV \sin(-120)) \\
 &= V + \frac{V}{2} + j\frac{\sqrt{3}V}{2} \\
 &= \frac{3V}{2} + j\frac{\sqrt{3}V}{2} \\
 &= \sqrt{3}V\angle 30^\circ \\
 &= \sqrt{3} V_{an} \angle 30^\circ
 \end{aligned}$$

By symmetry,

$$V_{bc} = \sqrt{3} V_{bn} \angle 30^\circ$$

$$V_{ca} = \sqrt{3} V_{cn} \angle 30^\circ$$

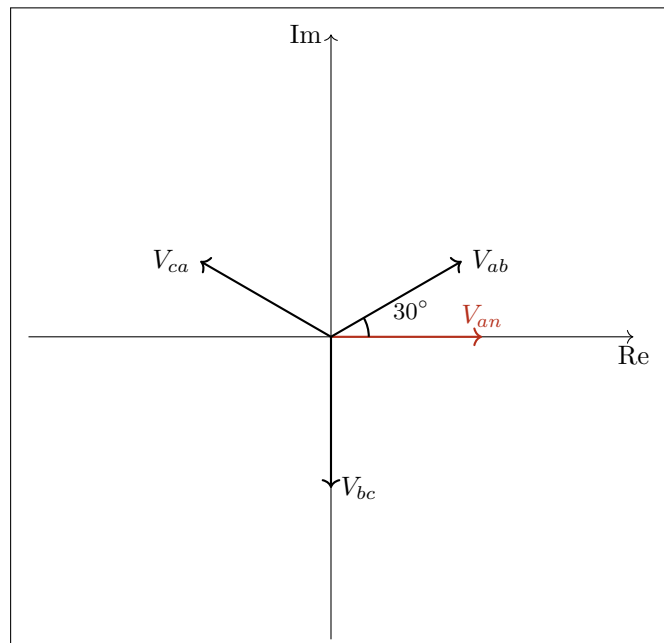


Figure 15: Phasor Diagram of line voltages

The phase difference between the line voltages and the phase voltages is both 120° , however, the line voltage is displaced by 30° .

In general,

$$V_L = \sqrt{3}V_P \angle 30^\circ \quad (5)$$

$$|V_L| = |V_{ab}| = |V_{bc}| = |V_{ca}|$$

$$I_L = I_P \quad (6)$$

In a Y connection, the line voltage is $\sqrt{3}$ times the phase voltage and leads the phase voltage by 30° . The line current is equal to the phase current.

6.4 Phase and line voltages in Delta connection

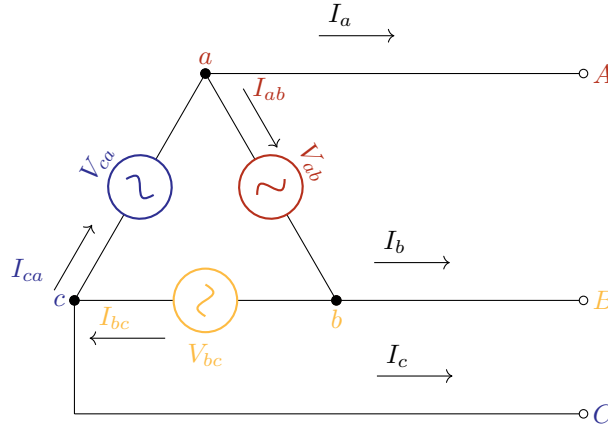


Figure 16: Three phase voltage source in Delta connection

$$V_L = V_P \quad (7)$$

Assuming impedances with zero phase and equal magnitude,

$$I_{ab} = \frac{V_{ab}}{Z} = \frac{V}{Z}$$

$$I_{bc} = \frac{V_{bc}}{Z} = \frac{V}{Z} \angle -120^\circ$$

$$I_{ca} = \frac{V_{ca}}{Z} = \frac{V}{Z} \angle 120^\circ$$

Using KCL,

$$\begin{aligned} I_a &= I_{ca} - I_{ab} \\ &= \frac{V}{Z} \angle 120^\circ - \frac{V}{Z} \\ &= \frac{V}{Z} \cos 120 + j \frac{V}{Z} \sin 120 - \frac{V}{Z} \\ &= \frac{-V}{2Z} + j \frac{\sqrt{3}V}{2Z} - \frac{V}{Z} \\ &= \frac{-3V}{2Z} + j \frac{\sqrt{3}V}{2Z} \\ &= \frac{\sqrt{3}V}{Z} \angle -30^\circ \\ &= \sqrt{3} I_{ab} \angle -30^\circ \end{aligned}$$

In general,

$$I_L = \sqrt{3} I_P \angle -30^\circ \quad (8)$$

In a Δ connection, the phase voltages and line voltages are the same. The line current is equal to $\sqrt{3}$ times the phase current and lags the phase current by 30°

6.5 Power in 3 Phase systems:

The total power in a three phase balanced system is equal to the sum of three equal phase powers.

$$P_{total} = 3P_P = 3V_P I_P \cos \theta \quad (9)$$

Here, θ is the load angle or the phase difference between current and voltage. It is easier to measure the line voltage and line current. So for a Y connection, using equations 5 and 6,

$$P_{total} = 3V_P I_P \cos \theta = 3 \frac{V_L}{\sqrt{3}} I_L \cos \theta = \sqrt{3} V_L I_L \cos \theta$$

For a Δ connection, using equations 7 and 8,

$$P_{total} = 3V_P I_P \cos \theta = 3V_L \frac{I_L}{\sqrt{3}} \cos \theta = \sqrt{3} V_L I_L \cos \theta$$

The same expression holds for both.

$$P_{total} = \sqrt{3} V_L I_L \cos \theta \quad (10)$$

$$Q_{reactive} = \sqrt{3} V_L I_L \sin \theta$$

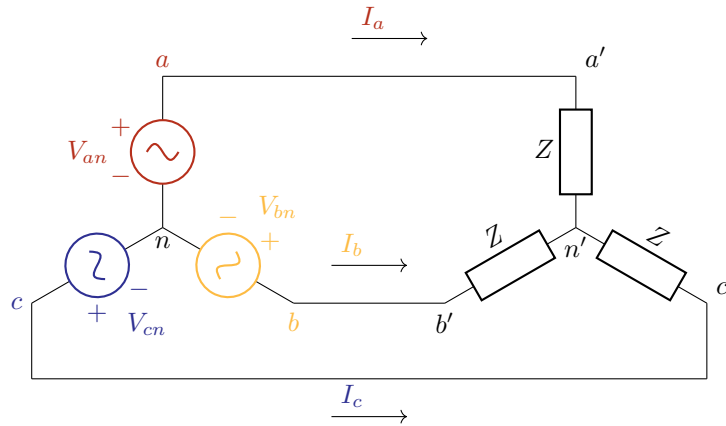
$$|S| = \sqrt{3} V_L I_L \quad (11)$$

6.6 Guidelines to solve questions:

- Draw a carefully oriented and clearly labeled wiring diagram.
- Analyze one phase and use symmetry for the others.
- Sketch a phasor diagram as a guide and for verification.

6.7 Example Questions

Question 1: 3 equal loads of impedance $Z = 10\angle 45^\circ$ are connected in Y across a 220 V supply. Find line and phase voltage, line and phase current, and phase and total power.



Solution: The source voltage represents the line voltage by default

$$V_L = 220V$$

$$V_L = \sqrt{3}V_P\angle 30^\circ$$

$$\Rightarrow V_P = \frac{V_L}{\sqrt{3}}\angle -30^\circ$$

$$V_{an} = \frac{220}{\sqrt{3}}\angle 0^\circ - 30^\circ$$

$$V_{an} = 127\angle -30^\circ V$$

By symmetry,

$$V_{bn} = 127\angle -30^\circ - 120^\circ$$

$$= 127\angle -150^\circ V$$

$$V_{cn} = 127\angle -30^\circ + 120^\circ$$

$$= 127\angle 90^\circ V$$

$$I_a = \frac{V_{an}}{Z} = \frac{127\angle -30^\circ}{10\angle 45^\circ} = 12.7\angle -75^\circ A$$

By symmetry,

$$I_b = 12.7 \angle -75^\circ - 120^\circ = 12.7 \angle -195^\circ A$$

$$I_c = 12.7 \angle -75^\circ + 120^\circ = 12.7 \angle 45^\circ A$$

Phase power is given by:

$$P_P = V_P I_P \cos \theta$$

$$= 127 \times \frac{12.7}{\sqrt{2}}$$

$$= 1140W$$

The expression for total power is:

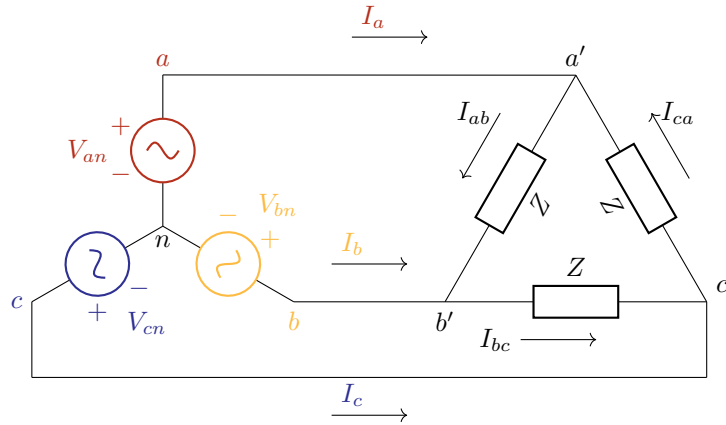
$$P_P = 3V_P I_P \cos \theta$$

$$= \sqrt{3} V_L I_L \cos \theta$$

$$= \sqrt{3} \times 220 \times \frac{12.7}{\sqrt{2}}$$

$$\approx 3420W$$

Question 2: 3 equal loads of impedance $Z = 10\angle 45^\circ$ are connected in Δ across a 220 V supply. Find line and phase voltage, line and phase current, and phase and total power.



Solution:

$$V_L = V_P = 220V$$

$$I_{ab} = \frac{V_{ab}}{Z} = \frac{220}{10\angle 45^\circ} = 22\angle -45^\circ A$$

By symmetry,

$$I_{bc} = 22\angle -45 - 120^\circ = 22\angle -165^\circ A$$

$$I_{bc} = 22\angle -45 + 120^\circ = 22\angle 75^\circ A$$

Line current $I_L = \sqrt{3}I_P\angle -30^\circ$

$$\begin{aligned} I_a &= \sqrt{3}I_{ab}\angle -30^\circ \\ &= 22\sqrt{3}\angle -45^\circ - 30^\circ \\ &= 38\angle -75^\circ A \end{aligned}$$

By symmetry,

$$I_b = 38\angle -75^\circ - 120^\circ = 38\angle -195^\circ A$$

$$I_c = 38\angle -75^\circ + 120^\circ = 38\angle 45^\circ A$$

$$P_P = V_P I_P \cos \theta$$

$$= 220 \times \frac{22}{\sqrt{2}}$$

$$= 3422W$$

$$\begin{aligned}P_{total} &= \sqrt{3}V_L I_L \cos \theta \\&= \sqrt{3} \times 220 \times \frac{38}{\sqrt{2}} \\&= 10238W\end{aligned}$$

In the above solved examples, it can be verified that total power is approximately thrice the phase power. Note that the power consumed in a Δ connection is approximately thrice of that consumed in a Y connection.

7 Contribution Statement

1. **TANVI PARSAM**- Complied all information on Latex for Lecture 18 and Lecture 19
2. **RAWAAN YOUSOF ALHAMMADI**- Made detailed notes for Lecture 18
3. **LATIFA TARIQ ALMARZOOQI**- Made detailed notes for Lecture 19 and 20
4. **SYEDA ESHAL ILHAN DAIMI**- Complied all information on Latex for Lecture 20